

A Pareto frontier between equilibrium fidelity and reversed-flow sign accuracy in neural wall function formulations

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Abstract

Wall-bounded turbulent flows under adverse pressure gradient (APG) and incipient separation violate the equilibrium assumptions of classical algebraic wall functions, yet the structural mechanism by which a near-wall closure represents reversed momentum flux is poorly understood. Within a single dataset and protocol—706 direct-numerical-simulation profiles from eight planar geometries spanning $\text{Re}_\tau \in [180, 5200]$ and five pressure-gradient regimes—we compare three nested closure depths: a Spalding-type algebraic baseline (Depth 0), a single-pass residual-correction network (Depth 1, DNCWF), and a non-equilibrium neural ordinary differential equation with a learned sign-unrestricted correction (Depth 2, FormNEQ). The results define a discrete Pareto frontier rather than a hierarchy. The Depth 1 closure cuts aggregate RMSE by 34% relative to Spalding (29% relative to the best per-band classical baseline) but *regresses* on canonical zero-pressure-gradient (−15%) and mild-APG (−19%) flow. A separation-point monotonicity lemma shows why an additively positive closure cannot represent reversed flow; the Depth 2 closure lifts this restriction and attains $99.4 \pm 1.0\%$ wall-stress-sign accuracy on reversed-flow profiles (three seeds) at the cost of three orders of magnitude in velocity RMSE, its learned correction overwhelming the equilibrium physics by up to $88\times$ near the wall. The deployment-mode wall-stress inversion is moreover fragile near separation

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(0% convergence in strong-APG cells). Equilibrium fidelity, reversed-flow sign accuracy, and inversion stability thus trade off structurally rather than tunably, constraining any neural wall function built on a single residual or single ODE family. Training data and code are released under a permanent DOI.

Keywords: turbulent boundary layer, wall function, flow separation, neural ordinary differential equation, adverse pressure gradient, Pareto frontier

1. Introduction

Wall-modelled computational fluid dynamics relies on wall functions to bridge the gap between the first off-wall grid point and the wall surface. The theoretical foundation is the law of the wall: in the inner region of a turbulent boundary layer, the velocity profile $U^+(y^+)$ follows a universal shape comprising a viscous sublayer ($U^+ = y^+$), a buffer layer, and a logarithmic region ($U^+ = \kappa^{-1} \ln y^+ + B$) [1, 2]. Spalding’s [3] implicit algebraic formula provides a smooth composite spanning all three sub-layers and remains the de facto standard in industrial CFD codes, alongside related formulations by Reichardt [4] and Werner & Wengle [5].

Under adverse pressure gradients (APG) and flow separation, the law of the wall breaks down. When the wall shear stress $\tau_w \rightarrow 0$ at separation, the friction velocity $u_\tau = \sqrt{|\tau_w|/\rho} \rightarrow 0$, and the inner scaling $y^+ = y u_\tau / \nu$ degenerates. Classical wall functions produce errors exceeding 100% of the reference wall stress in separated regions [6], severely degrading the predictive capability of wall-modelled LES and RANS computations for engineering flows involving separation.

Machine learning has emerged as a route to construct wall closures that extend beyond the log-law regime. The recent literature spans several architectural families. Building-block-flow models [7] explicitly handle separated and reattaching flows by classifying the local regime and dispatching to a regime-specific predictor; the Lozano-Durán–Bae closure was deployed a-posteriori in WMLES and demonstrated sign-correct behaviour in separated regions.¹ Non-equilibrium ML wall closures with embedded streamwise

¹We follow turbulence-modelling convention: *a-priori* testing evaluates a closure on fixed high-fidelity (DNS) inputs in isolation, with no feedback into a flow solver, whereas *a-posteriori* testing embeds the closure inside a coupled solver (wall-modelled LES or

pressure-gradient source terms have been advanced by Bin and co-workers [8] for separated channels and reversed-flow regimes. A neural wall function with adjoint-based sensitivity and a-posteriori LES validation was reported by Dupuy and co-workers [9] on flat-plate and APG TBLs. A recent survey catalogues these supervised, reinforcement-learning, and memory-based wall-model families and their relative trade-offs [10]. Reinforcement-learning wall closures have also been demonstrated for high-Reynolds-number channel flow [11]. Profile-fitting and PINN-based approaches [12] sit at adjacent points on the same architectural spectrum.

A consequence of this rapid expansion is that each existing study uses different training data, evaluation metrics, baseline treatments, and deployment loops, so the architectural choice itself is confounded with the experimental setup. We do *not* claim that any of the closure families above is unexplored; reversed-flow detection in particular has been addressed in earlier neural wall closures. What is missing in the literature is a controlled comparison *within a single dataset and protocol* of how the depth of physics structure interacts with the two failure modes that dominate the application: equilibrium accuracy on attached flow, and sign accuracy on reversed flow.

This paper presents that controlled comparison. We define *formulation depth* as the degree of physics structure embedded in a neural wall function, and evaluate three nested depths on a single dataset, single protocol, and single feature vector:

- Depth 0. Classical algebraic wall functions** (Spalding, Reichardt, Werner–Wengle): closed-form expressions with no learnable parameters.
- Depth 1. Direct Neural Correction Wall Function (DNCWF)**: a single-forward-pass architecture that adds a learned residual to Spalding’s law, with structural no-slip enforcement and asymptotic-preserving blending (10,845 parameters).
- Depth 2. Non-equilibrium Neural ODE (FormNEQ)**: the learned closure of the governing boundary-layer ODE, integrating a physics-structured equilibrium term with a learned non-equilibrium correction via a fixed-step ODE solver (12,254 parameters).

The gap between Depth 0 and Depth 1 measures the *value of learning*:

RANS) and assesses the resulting simulation. The distinction matters because a closure that is accurate a-priori can still destabilise or bias an a-posteriori run. All quantitative results in this paper are a-priori.

how much does a neural correction improve over closed-form laws? The gap between Depth 1 and Depth 2 measures the *value of ODE structure beyond learning*: does embedding the governing differential equation provide qualitative capabilities—such as reversed-flow detection—that justify the computational overhead of numerical integration?

All learned formulations are trained on the same 706 DNS profiles from 8 geometries spanning 5 pressure-gradient regimes in Duprat’s [6] hybrid velocity coordinates, using the same 6-dimensional feature vector, the same optimizer and hyperparameters, the same loss function, and the same evaluation metrics. The *only* independent variable is the formulation depth.

The results define a Pareto frontier rather than a hierarchy.² Depth 1 (DNCWF) reduces aggregate RMSE by 34% relative to Spalding—or by 29% relative to the best per-band classical baseline (Section 5.2)—and generalises well within the periodic-hill family ($< 5\%$ training–holdout gap), but *regresses* on canonical ZPG flow (-15% vs Reichardt) and mild APG (-19% vs Spalding); equilibrium accuracy is sacrificed for adverse-gradient correction. Depth 2 (FormNEQ) reaches $99.4 \pm 1.0\%$ reversed-flow (R5) sign accuracy across three seeds at the cost of three orders of magnitude in velocity RMSE, with a g_θ /equilibrium dominance ratio of $88\times$ near the wall. A separation-point monotonicity lemma (Section 3.2) explains *why* a Depth 1 closure with non-negative coefficients cannot leave $\tau_w \geq 0$ at the boundary; the lemma does not extend to $\tau_w < 0$, where reversed-flow representability becomes an architectural choice (sign-restricted vs sign-unrestricted) rather than a structural impossibility. For deployment in a coupled solver loop, the DNCWF closure does not require an ODE inversion (it is a single-pass forward model), whereas the FormNEQ closure does; we do not evaluate FormNEQ’s a-posteriori inversion stability here and treat ODE-form inversion as a baseline-level deployment risk, quantified in Section 6.4.

The contributions of this paper are:

²We use *Pareto frontier* in its precise sense: the set of closures that are not dominated by any other evaluated closure on the two objectives of interest (aggregate velocity RMSE and reversed-flow sign accuracy). Among the closures tested this non-dominated set is the discrete two-point set {DNCWF, FormNEQ}—DNCWF minimises RMSE, FormNEQ maximises sign accuracy, and the classical baselines are dominated by DNCWF. We therefore mean a discrete non-dominated set of operating points, not a continuous frontier traced by sweeping a one-parameter family of architectures; characterising such a continuum is left to future work (Section 6.4).

1. A controlled, within-protocol comparison of three neural wall function formulation depths (algebraic baselines; single-pass residual; non-equilibrium Neural ODE) on a single 706-profile DNS dataset, with identical Duprat hybrid coordinates, 6-dimensional feature vector, optimiser, and evaluation metric.
2. Quantitative evidence of a Pareto frontier between equilibrium fidelity, reversed-flow sign accuracy, and inversion stability: no single closure dominates across the five regime bands, and the trade-offs are reproducible in our experiments.
3. A separation-point monotonicity lemma (Theorem 1) that establishes non-decreasing $U^*(y^*)$ at $\tau_w = 0$ for Depth 1 closures with non-negative coefficients in Duprat scaling, and a corollary identifying the inner-coordinate region in which mild reversed flow can be represented for $\tau_w < 0$.
4. A g_θ dominance-ratio diagnostic that decomposes the FormNEQ dynamics into equilibrium and non-equilibrium components, demonstrating that the learned correction is pathologically constant ($< 15\%$ variation across $y^* \in [1, 300]$) and overwhelms the equilibrium term by up to $88\times$ near the wall—identifying the integral-momentum constraint as the missing ingredient.
5. A diagnostic verification that the asymptotic-preserving α gate, parameterised in β_{Clauser} , recovers regime detection consistent with Duprat’s hybrid scaling across both learned formulations.

1.1. Scope and history of this study

We disclose the scope contraction of this work, since it bears on the interpretation of the contributions above. The study began as an attempt to train a single *universal* neural wall function on the full 8-geometry corpus and to demonstrate dominance over Spalding across all five regime bands, all training geometries, and an extended Reynolds-number range. The 8-geometry vanilla MLP closure trained under that scope reached an aggregate RMSE of 22.4 (31.7% *worse* than Spalding’s 17.0 on the same data), and we were unable to close the universality gates within the architectural family attempted. We pivoted from a universality target to a controlled within-protocol comparison of formulation depths, which is what the paper presents. The Pareto-frontier framing is therefore a finding, not a hypothesis we set out to confirm; the negative result on universality is, on our evidence, structural

rather than an artefact of training. We discuss in Section 6.4 the experimental package required to either close the universality gates or reject them in a stronger sense than reported here.

2. Background

2.1. Classical wall function theory

The near-wall velocity field is governed by the thin-shear-layer equation. Once-integrating in the wall-normal direction from $y = 0$ yields

$$(\nu + \nu_t) \frac{dU}{dy} = \frac{\tau_w}{\rho} + \frac{1}{\rho} \frac{dp}{dx} y, \quad (1)$$

in the equilibrium limit (convective terms neglected). With $\nu_t = 0$, equation (1) yields the viscous sublayer $U^+ = y^+$. With the mixing-length model $\nu_t = \kappa^2 y^2 |dU/dy|$ and Van Driest damping $\ell = \kappa y [1 - \exp(-y^+/A^+)]$, one obtains the smooth transition through the buffer layer to the logarithmic region [13]. Spalding's [3] implicit algebraic formula

$$y^+ = U^+ + e^{-\kappa B} \left[e^{\kappa U^+} - 1 - \kappa U^+ - \frac{(\kappa U^+)^2}{2} - \frac{(\kappa U^+)^3}{6} \right], \quad (2)$$

with $\kappa = 0.41$ and $B = 5.2$, provides a continuous profile from the wall to the log layer in a single expression. Reichardt [4] proposed an explicit three-layer formula that avoids the implicit inversion required by Spalding but shares the same equilibrium assumptions. Werner & Wengle [5] developed a power-law approximation that is computationally cheaper for LES time-stepping.

All classical wall functions assume $\tau_w > 0$ and use friction-velocity scaling $y^+ = y u_\tau / \nu$. Under adverse pressure gradients, τ_w decreases and the log-law deviates; at separation, $\tau_w \rightarrow 0$ and the scaling becomes singular. Duprat et al. [6] resolved this by introducing the hybrid velocity scale

$$u_{\tau p}^3 = u_\tau^3 + \alpha u_p^3, \quad u_p = \left(\frac{\nu}{\rho} \left| \frac{dp}{dx} \right| \right)^{1/3}, \quad (3)$$

with $\alpha = 1$ [6], which remains finite at separation, enabling wall-normal scaling $y^* = y u_{\tau p} / \nu$ and velocity scaling $U^* = U / u_{\tau p}$ across all pressure-gradient regimes.

2.2. Machine-learned wall functions

Data-driven wall functions can be grouped into three categories.

Profile-fitting models. Yang et al. [12] trained a physics-informed neural network to predict the inner-scaled velocity profile $U^+(y^+)$ from local flow features as a wall model for LES. These approaches bypass the ODE structure of the boundary layer and enforce $U^+(0) = 0$ through a penalty term.

Wall-stress predictors. Zhou et al. [14] used an artificial neural network to construct a subgrid-scale closure from filtered velocity features; Lozano-Durán et al. [7] developed building-block models trained on channel flow DNS. These provide scalar wall stress without profile reconstruction or interpretable closures.

Physics-informed approaches. Kawai and Larsson [15] and Park and Moin [16] solved non-equilibrium and equilibrium thin-boundary-layer equations on the LES grid, using analytical closures.

No prior work formulates the wall function as a Neural ODE that learns the closure of the governing boundary-layer ODE. All existing ML approaches either fit the profile directly (bypassing the ODE), predict a scalar output (τ_w), or embed physics as soft loss constraints (PINNs) rather than as the computational structure.

2.3. Neural ordinary differential equations

Neural ODEs [17] parameterise the right-hand side of an ODE by a neural network and compute gradients via the continuous adjoint method with $O(1)$ memory cost. Universal differential equations (UDEs) [18] extend this framework to learn unknown terms within known differential equations—precisely the paradigm of the present work. Software implementations include `torchdiffeq` [19] for PyTorch and `diffjax` [20] for JAX. Neural ODEs have been applied to various physical systems [18], but not to turbulent wall functions.

3. Methodology

3.1. Problem statement: the formulation depth spectrum

Equation (1) in Duprat coordinates (3) (equilibrium limit) becomes the initial-value problem

$$\left(1 + \frac{\nu_t}{\nu}\right) \frac{dU^*}{dy^*} = C_\tau + C_p y^*, \quad U^*(0) = 0, \quad (4)$$

where the source coefficients

$$C_\tau = \left(\frac{u_\tau}{u_{\tau p}} \right)^2 \text{sign}(u_\tau), \quad C_p = \left(\frac{u_p}{u_{\tau p}} \right)^3 \quad (5)$$

satisfy $C_\tau \rightarrow 1$, $C_p \rightarrow 0$ at zero pressure gradient and $C_\tau \rightarrow 0$, $C_p \rightarrow 1$ at separation.

A neural wall function must choose *how much* of this physics to embed. At one extreme, the boundary-layer ODE (4) is prescribed and only the closure ν_t is learned (maximum physics structure). At the other extreme, the profile $U^*(y^*)$ is predicted directly with no differential-equation structure (maximum flexibility). We call this choice the *formulation depth*.

This paper compares three formulation depths under controlled conditions:

- **Depth 0:** Classical algebraic wall functions—closed-form expressions with zero learnable parameters (Section 2.1).
- **Depth 1:** The DNCWF—a direct neural correction to Spalding, evaluated in a single forward pass with no ODE solver (Section 3.4).
- **Depth 2:** FormNEQ—a non-equilibrium Neural ODE that learns both the eddy viscosity and a convective correction, integrated via a fixed-step ODE solver (Section 3.3).

The gap between Depth 0 and Depth 1 isolates the value of *learning from data*. The gap between Depth 1 and Depth 2 isolates the value of *embedding the governing ODE* beyond learning alone. Before presenting these formulations, we first establish a structural limitation shared by all equilibrium ODE formulations (Section 3.2), which motivates the non-equilibrium correction in FormNEQ.

3.2. Separation-point monotonicity and the inner-coordinate reach of mild reversed flow

Before presenting the two learned formulations (Depth 1 and Depth 2), we establish a precise statement about what an equilibrium-form ODE closure can and cannot represent at and near the separation point. The equilibrium thin-shear-layer ODE (4) admits the equivalent form

$$\frac{dU^*}{dy^*} = \frac{S^*(y^*)}{1 + \nu_t(y^*)/\nu}, \quad U^*(0) = 0, \quad (6)$$

for any non-negative eddy viscosity profile $\nu_t(y^*) \geq 0$.

Theorem 1 (Separation-point monotonicity). *At the separation point $\tau_w = 0$ with $dp/dx \neq 0$, every solution of (6) with $\nu_t \geq 0$ satisfies $U^*(y^*) \geq 0$ for all $y^* \geq 0$.*

Proof. At $\tau_w = 0$, the friction velocity $u_\tau = 0$. The Duprat mixed scale (3) reduces to $u_{\tau p} = \alpha^{1/3} u_p > 0$ (since $dp/dx \neq 0$ implies $u_p > 0$). The source coefficients (5) become

$$C_\tau = \left(\frac{0}{u_{\tau p}} \right)^2 \cdot \text{sign}(0) = 0, \quad C_p = \left(\frac{u_p}{\alpha^{1/3} u_p} \right)^3 = \frac{1}{\alpha} > 0.$$

The source term $S^*(y^*) = C_p y^* \geq 0$ for $y^* \geq 0$, and the denominator $1 + \nu_t/\nu \geq 1 > 0$. Therefore $dU^*/dy^* \geq 0$, and integration from $U^*(0) = 0$ yields $U^*(y^*) \geq 0$. $\square$

Scope of Theorem 1. We emphasise that the theorem is a statement about the boundary case $\tau_w = 0$ only. It is *not* a no-go theorem for reversed-flow representability under $\tau_w < 0$: in that regime the source coefficient $C_\tau = (\tau_w/u_{\tau p}^2) \text{sign}(\tau_w) < 0$, the source $S^*(y^*) = C_\tau + C_p y^*$ is negative near the wall and positive farther out, and the resulting solution can be non-monotone with a near-wall reversed-flow lobe (Corollary 2). The relevant architectural choice is therefore not whether an equilibrium-form ODE can represent reversed flow—it can, in principle, when $\tau_w < 0$ —but whether the closure constrains its source coefficients to remain non-negative. A Depth 1 residual built additively on a positive-definite Spalding baseline (DNCWF, Section 3.4) cannot produce $U^* < 0$ regardless of the underlying τ_w , because the construction forces $U^* \geq 0$ everywhere. A closure that parameterises C_τ without a sign restriction *can*, in principle, represent reversed flow in the $\tau_w < 0$ regime; whether it does so reliably is then an empirical question that this paper addresses for the two specific architectures evaluated.

Corollary 2 (Inner-coordinate reach of mild reversed flow). *For mild reverse flow ($\tau_w < 0$, $|\tau_w| \ll \rho u_p^2$), the source term S^* changes sign at $y_0^* = |C_\tau|/C_p$, confining the reverse-flow region to $y^* \in [0, y_0^*]$. As $\tau_w \rightarrow 0^-$, $|C_\tau| \rightarrow 0$ and $y_0^* \rightarrow 0$: the reverse-flow signal collapses into an infinitesimally thin layer in Duprat coordinates.*

Numerical check on R5 DNS profiles.. We evaluated the corollary directly on the 205 reversed-flow (R5) training profiles for which $C_p > 0$. The predicted equilibrium-source sign-change location $y_0^* = |C_\tau|/C_p$ has a median of 0.008 and a maximum of 0.48 across these profiles, and shrinks systematically as $\tau_w \rightarrow 0^-$, confirming the predicted collapse of the equilibrium reverse-flow signal into a near-wall layer of vanishing thickness in Duprat coordinates. The actual reversed-flow region in the DNS, by contrast, is thick: the inner-scaled velocity reaches its near-wall minimum at a median $y^* \approx 105$ and returns to $U^* = 0$ at a median $y^* \approx 317$. The two scales differ by several orders of magnitude because the corollary is an *equilibrium-source* statement (it neglects the eddy-viscosity and convective history terms), whereas the DNS reverse flow is sustained by exactly those non-equilibrium terms. This is the quantitative content of the representability gap: an equilibrium-form closure predicts an infinitesimally thin reverse layer, so it cannot reproduce the observed thick reverse flow—precisely the deficit that the non-equilibrium correction in FormNEQ (Section 3.3) is introduced to supply.

This inner-coordinate contraction is one mechanism behind the empirically observed 0% reversed-flow sign accuracy of Depth 0 and Depth 1 closures in our experiments (Section 5.3): classical Depth 0 baselines assume $\tau_w > 0$, and the DNCWF (Depth 1) inherits an additive construction on a positive-definite Spalding baseline. The Depth 2 FormNEQ closure (Section 3.3) introduces a sign-unrestricted non-equilibrium term that can, in principle, lift this restriction at the cost of magnitude stability; the rest of the paper quantifies the resulting trade-off.

3.3. Depth 2: Non-equilibrium Neural ODE (FormNEQ)

FormNEQ (Depth 2) is the deepest formulation in our taxonomy: it learns the closure of the governing boundary-layer ODE through numerical integration. The non-equilibrium correction g_θ introduced below is sign-unrestricted, so the architecture is not subject to the additive-positivity restriction that the DNCWF inherits from its Spalding baseline. FormNEQ addresses this by restoring the convective acceleration terms from the full thin-shear-layer equation:

$$U \frac{\partial U}{\partial x} + V \frac{\partial U}{\partial y} = -\frac{1}{\rho} \frac{dp}{dx} + \frac{\partial}{\partial y} \left[(\nu + \nu_t) \frac{\partial U}{\partial y} \right]. \quad (7)$$

In separated flow these convective terms are large: the streamwise velocity decelerates rapidly and the wall-normal velocity V is comparable to U . Form-

NEQ restores these terms approximately through a learned non-equilibrium correction:

$$\frac{dU^*}{dy^*} = \underbrace{\frac{S^*(y^*)}{1 + \nu_{t,\theta}(y^*, \boldsymbol{\varphi})/\nu}}_{\text{equilibrium term}} + \underbrace{\alpha(\beta) g_\theta(y^*, U^*, \boldsymbol{\varphi}, \mathbf{g})}_{\text{non-equilibrium correction}}, \quad U^*(0) = 0, \quad (8)$$

where:

- $\nu_{t,\theta}$ is a Softplus-enforced eddy viscosity network;
- $\alpha(\beta)$ is the fixed asymptotic-preserving blending (11), with $\alpha(0) = 0$ (inactive in ZPG) and $\alpha(\beta \rightarrow \infty) \rightarrow 1$ (fully active at separation);
- g_θ is an unconstrained FiLM-conditioned network (the *NEQ correction network*) with near-zero output initialisation;
- $\mathbf{g} \in \mathbb{R}^4$ is the ContinuousGeoEncoder embedding (13).

The critical structural difference from all equilibrium formulations: g_θ is *unconstrained* and can produce negative values. When $\alpha \approx 1$ and $g_\theta < 0$ with $|g_\theta| > S^*/(1 + \nu_t/\nu)$, the total gradient $dU^*/dy^* < 0$, breaking the monotonicity of Theorem 1 and enabling the ODE to represent reverse flow.

Structural properties.. FormNEQ inherits exact no-slip enforcement ($U^*(0) = 0$ as initial condition) and physics-informed ODE structure from the equilibrium boundary-layer equation. It reduces exactly to the equilibrium ODE when $\alpha = 0$ (attached flow), ensuring that R1/R2 accuracy is preserved by construction—not by training. The fixed $\alpha(\beta)$ avoids gate-collapse failure modes: unlike a learned gate that could converge to $\sigma \approx 0$ uniformly, α is a fixed function of the Clauser parameter and cannot be suppressed by the optimizer. Near-zero initialisation of g_θ ensures that the optimisation starts from the equilibrium solution, providing a stable starting point.

NEQ correction network architecture.. The network $g_\theta : \mathbb{R}^{2+K} \times \mathbb{R}^{d_g} \rightarrow \mathbb{R}$ takes $(y^*, U^*, \boldsymbol{\varphi})$ as input and produces a scalar correction, conditioned on the geometry embedding $\mathbf{g}$ via FiLM layers. It comprises three hidden layers of width 64 with tanh activations and FiLM conditioning, with output-layer weights initialised at $\mathcal{O}(10^{-2})$ and bias at zero. Unlike the eddy-viscosity network $\nu_{t,\theta}$ (which omits U^* to prevent ODE stiffness), g_θ requires U^* as input because the convective correction depends on the local velocity state.

Algorithm 1 FormNEQ two-phase training with equilibrium warmup

Require: Training profiles $\mathcal{D} = \{(y_j^*, U_j^*, \varphi_j, C_{\tau,j}, C_{p,j})\}_{j=1}^N$

Require: Total epochs E , warmup epochs W , regularisation weight λ_g

```
1: Initialise FormNEQ with  $g_\theta$  output weights  $\sim \mathcal{O}(10^{-2})$ 
2: for  $e = 1$  to  $E$  do
3:   if  $e \leq W$  then
4:     Freeze  $g_\theta$  parameters ▷ Phase 1: equilibrium only
5:   else if  $e = W + 1$  then
6:     Unfreeze  $g_\theta$  parameters ▷ Phase 2: full training
7:   end if
8:   for each profile  $j \in \{1, \dots, N\}$  do
9:     Set  $C_\tau, C_p, \alpha(\beta_j)$  via set_profile_params
10:     $\hat{U}_j^* \leftarrow \text{odeint}(\text{FormNEQ}, U^*(0)=0, y_j^*)$ 
11:     $\mathcal{L}_j \leftarrow \|\hat{U}_j^* - U_j^*\|^2$ 
12:    if  $e > W$  then
13:       $\mathcal{L}_j \leftarrow \mathcal{L}_j + \lambda_g \|g_\theta(y_j^*, \hat{U}_j^*, \varphi_j, \mathbf{g}_j)\|^2$ 
14:    end if
15:  end for
16:   $\nabla \leftarrow \nabla_\theta \sum_j \mathcal{L}_j$ ; clip  $\|\nabla\| \leq 1$ ; update  $\theta$ 
17: end for
18: return trained model
```

Training protocol. To prevent the correction from destabilising the equilibrium base, training proceeds in two phases (Algorithm 1): (i) *equilibrium warmup* (first W epochs): g_θ is frozen and only $\nu_{t,\theta}$ is trained, establishing a stable equilibrium solution; (ii) *full training* (remaining epochs): both networks are unfrozen, with an L^2 penalty $\lambda_g \|g_\theta\|^2$ on the correction magnitude to keep the non-equilibrium term small where it is not needed. The warmup ensures the optimiser starts from the equilibrium solution, providing a stable basin from which the non-equilibrium correction can be gradually introduced.

3.4. Depth 1: Direct Neural Correction Wall Function (DNCWF)

The DNCWF (Depth 1) represents the intermediate formulation depth: it adds a learned correction to a classical baseline without requiring ODE integration. The velocity profile is predicted in a single forward pass by

adding a structurally constrained neural correction to Spalding’s law:

$$U^*(y^*) = U_{\text{Spalding}}^*(y^*) + \alpha(\beta) \frac{y^*}{y_{\text{max}}^*} \Delta_\theta \left(\frac{y^*}{y_{\text{max}}^*}, \boldsymbol{\varphi}, \mathbf{g} \right), \quad (9)$$

where U_{Spalding}^* is Spalding’s law (2) evaluated in Duprat coordinates, $\alpha(\beta)$ is a fixed asymptotic-preserving blending function, y^*/y_{max}^* is a structural no-slip multiplier, and Δ_θ is a learned neural correction conditioned on physics features $\boldsymbol{\varphi}$ and geometry embedding $\mathbf{g}$.

3.5. Spalding baseline

Spalding’s implicit law (2) is inverted numerically to obtain $U^+(y^+)$ via batched Newton–Raphson iteration:

$$U_{n+1}^+ = U_n^+ - \frac{y^+(U_n^+) - y^*}{(\mathrm{d}y^+/\mathrm{d}U^+)|_{U_n^+}}, \quad n = 0, 1, \dots, 19, \quad (10)$$

initialised from the log-law approximation $U_0^+ = \kappa^{-1} \ln \max(y^*, 0.5) + B$. The Spalding evaluation is *detached* from the computational graph: gradients flow only through the neural correction Δ_θ , not through the baseline.

3.6. Asymptotic-preserving blending

The blending function α controls correction strength based on the Clauser equilibrium parameter $\beta = \beta_{\text{Clauser}}$:

$$\alpha(\beta) = \max(0, \sigma(k(\beta - \beta_0)) - \sigma(-k\beta_0)), \quad (11)$$

where $\sigma(x) = (1 + e^{-x})^{-1}$ is the logistic sigmoid, and $\beta_0 = 1.0$ centres the transition. The steepness parameter k controls the transition sharpness: we use $k = 3.0$ for DNCWF and $k = 5.0$ for FormNEQ.

Properties of $\alpha(\beta)$.

- $\alpha(0) = \max(0, \sigma(-k\beta_0) - \sigma(-k\beta_0)) = 0$ *exactly*, for any k and β_0 .
- $\alpha(\beta) > 0$ for all $\beta > 0$, and α is monotonically increasing.
- $\alpha(\beta \rightarrow \infty) \rightarrow 1 - \sigma(-k\beta_0) \approx 0.993$ for $k = 5$, $\beta_0 = 1$.
- α is C^∞ everywhere—no gradient discontinuities.

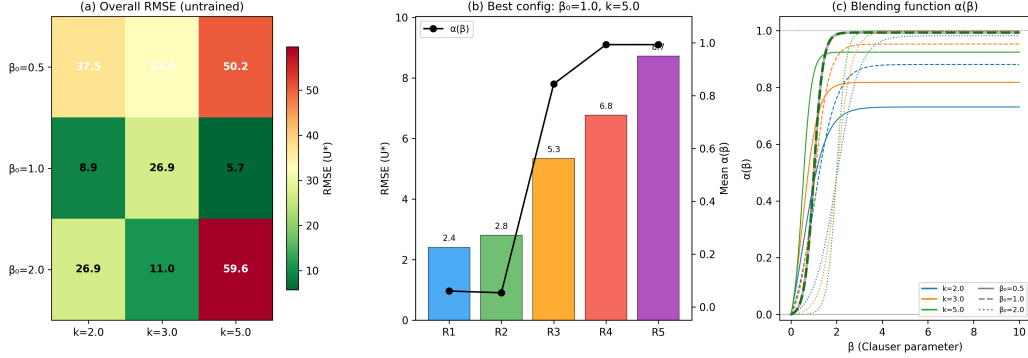


Figure 1: Sensitivity of the AP blending function to β_0 and k . (a) Overall RMSE (untrained model) across 9 configurations. (b) Per-regime RMSE and mean $\alpha(\beta)$ for representative configurations. (c) Blending function $\alpha(\beta)$ curves for all configurations.

- α is *not* learned: it is a fixed function computed from the input β_{Clauser} .

The fixed blending eliminates gate-collapse failure modes: a learned gate $\sigma_\theta(\boldsymbol{\varphi})$ can converge to $\sigma \approx 0$ uniformly across all regimes, zeroing the gradient to the correction network regardless of prediction error. By fixing α as a function of the physically meaningful β_{Clauser} , the correction network receives non-zero gradient signal for all profiles with $\beta > 0$.

Fixed reference length versus the local Clauser δ^ .* The gate is driven by β_{Clauser} , which uses a fixed reference length $\delta_{\text{ref}} = 0.1$ m rather than the local displacement thickness δ^* of the Clauser parameter. We adopt the fixed scale for two reasons: it is robust in the separated regimes, where δ^* is numerically ill-defined as the edge velocity $U_e \rightarrow 0$; and, because the gate then depends on the profile only through $|dp/dx|/|\tau_w|$, it cannot be destabilised by a noisy integral of a reversed profile. To quantify the effect of the choice we recomputed β_{Clauser} with the local δ^* on every training profile and passed both through the gate (Table 1, FormNEQ steepness $k = 5$). The binary gating pattern—near-zero in equilibrium (R1/R2), near-one in non-equilibrium (R3–R5)—is preserved under both length scales, and R4/R5 are essentially unchanged (the gate is saturated there). The one material difference is the mild-APG band R2, where the local δ^* raises the mean activation from 0.07 to 0.50. Because R2 is precisely the band on which the learned correction already *regresses* (Section 5.1), the larger δ^* -driven

Table 1: Mean gate activation $\alpha(\beta_{\text{Clauser}})$ per regime (FormNEQ steepness $k = 5$) under the fixed reference length $\delta_{\text{ref}} = 0.1$ m used in this paper versus the local Clauser displacement thickness δ^* . The binary equilibrium/non-equilibrium pattern is preserved; only the mild-APG band R2 changes materially.

	R1	R2	R3	R4	R5
$\delta_{\text{ref}} = 0.1$ m (used)	0.059	0.070	0.908	0.993	0.982
local δ^*	0.010	0.497	0.993	0.993	0.962

activation would inject more correction where it is least helpful; the fixed δ_{ref} is the more conservative choice for protecting equilibrium accuracy. The regime ordering and the qualitative gating are unaffected by the choice.

3.7. Structural no-slip enforcement

The multiplicative factor y^*/y_{max}^* in equation (9) enforces the no-slip condition by construction:

$$U^*(0) = U_{\text{Spalding}}^*(0) + \alpha(\beta) \cdot \frac{0}{y_{\text{max}}^*} \cdot \Delta_{\theta}(\dots) = 0 + 0 = 0, \quad (12)$$

since $U_{\text{Spalding}}^*(0) = 0$ by Spalding’s law. This is a *structural* guarantee: no learned parameter, training epoch, or weight initialisation can violate it.

3.8. FiLM-conditioned correction network

The correction network Δ_{θ} uses Feature-wise Linear Modulation (FiLM; [21]) for geometry-aware conditioning.

Geometry encoder.. The *ContinuousGeoEncoder* maps the feature vector to a compact geometry embedding:

$$\mathbf{g} = \tanh(W_g \boldsymbol{\varphi} + b_g) \in \mathbb{R}^4. \quad (13)$$

Because $\mathbf{g}$ is a continuous function of $\boldsymbol{\varphi}$ (not a discrete look-up table), the encoder accepts any feature vector at inference—including from geometries never seen during training—enabling zero-shot geometric generalisation.

FiLM modulation.. After each hidden layer, the representation is modulated by geometry-dependent affine parameters:

$$h' = \gamma(\mathbf{g}) \odot h + \beta(\mathbf{g}), \quad (\gamma, \beta) = W_{\text{FiLM}} \mathbf{g} + b_{\text{FiLM}}, \quad (14)$$

where h is the hidden representation and $\odot$ denotes element-wise multiplication.

Network architecture. The DNCWF correction network receives $[y^*/y_{\max}^*, \boldsymbol{\varphi}] \in \mathbb{R}^7$ and consists of three hidden layers (width 64, tanh activations), each followed by a FiLM layer (14), and a scalar output layer. The output weights are scaled by 0.01 and biases set to zero, so $\Delta_\theta \approx 0$ at the start of training. Combined with $\alpha(0) = 0$ at zero pressure gradient, the initial model is exactly Spalding’s law—training begins from a known-good baseline and learns corrections incrementally. The total parameter count is 10,845.

3.9. Structural properties of the DNCWF

Four structural properties are guaranteed by the DNCWF architecture:

- P1. Exact Spalding recovery at ZPG.** For any profile with $\beta_{\text{Clauser}} = 0$: $\alpha(0) = 0$ exactly by equation (11), so $U^*(y^*) = U_{\text{Spalding}}^*(y^*)$ for all y^* , independent of network weights θ .
- P2. Exact no-slip condition.** For any features $\boldsymbol{\varphi}$ and any weights θ : $U^*(0) = 0$ exactly by equation (12).
- P3. No gate collapse.** Since α is fixed (not learned), for all $\beta > 0$: $\alpha(\beta) > 0$, and the gradient $\partial \mathcal{L} / \partial \Delta_\theta \propto \alpha(\beta) \neq 0$ whenever the prediction error is non-zero.
- P4. $O(1)$ forward pass.** The prediction cost is $O(n \cdot d^2)$ where n is the number of grid points and $d = 64$ is the hidden dimension, with no sequential ODE integration steps.

3.10. Physics-based input features

Both learned formulations (DNCWF and FormNEQ) receive $K = 6$ non-dimensional physics features as auxiliary inputs:

$$\boldsymbol{\varphi} = \left[\ln(\text{Re}_\tau + 1), p^+, \beta_{\text{Clauser}}, \hat{C}_p, s_{\tau_w}, \hat{V} \right], \quad (15)$$

where the individual features are:

- $\ln(\text{Re}_\tau + 1)$: turbulence maturity, encoding the Reynolds-number dependence;
- $p^+ = (dp/dx) \nu / (\rho u_\tau^3)$: the inner-scaled dimensionless pressure gradient, clipped to $[-100, 100]$;
- $\beta_{\text{Clauser}} = |dp/dx| \delta_{\text{ref}} / \max(|\tau_w|, 10^{-12})$: the Clauser [22] adverse-pressure-gradient strength parameter, with a fixed reference length scale $\delta_{\text{ref}} = 0.1$ m in place of the local displacement thickness (the choice is examined in Section 3.6), clipped to $[0, 100]$;

- $\hat{C}_p = C_p/(|C_\tau| + |C_p| + \varepsilon)$: the relative pressure-gradient strength in Duprat source coefficients;
- $s_{\tau_w} = \tanh(\tau_w/(|\tau_w| + 10^{-10}))$: a continuous wall-shear-stress sign indicator;
- $\hat{V} = \max |V|/\max |U|$: the wall-normal-to-streamwise velocity ratio, a non-equilibrium indicator, clipped to $[0, 10]$.

3.11. Training procedure

Both learned formulations share the same training protocol to ensure a fair comparison. At training time, τ_w is known (from DNS), so u_τ , u_p , and $u_{\tau p}$ are computed directly. Each profile $\{y_i, U_i\}_{i=1}^N$ is converted to Duprat coordinates $\{y_i^*, U_i^*\}$. During training, the grid is subsampled to $N_{\text{train}} = 20$ points per profile. The DNCWF evaluates equation (9) in a single vectorised forward pass, while FormNEQ integrates equation (8) via forward Euler on the same 20-point grid.

Regime-weighted loss.. The base loss for a single profile is the mean squared error in U^* coordinates:

$$\ell_k = \frac{1}{N_k} \sum_{i=1}^{N_k} (U_{\text{pred}}^*(y_{k,i}^*) - U_{\text{true}}^*(y_{k,i}^*))^2. \quad (16)$$

Profiles are assigned to one of five pressure-gradient regimes (Section 3.12), and regime weights approximately follow inverse frequency to ensure under-represented regimes receive proportionally higher gradient signal.

Matching-height augmentation (DNCWF only).. In deployment, a wall function is evaluated at the first off-wall grid point, whose y^+ varies with mesh resolution. To reduce sensitivity to the matching height, each DNCWF training profile contributes losses at multiple y_{match}^+ cutoffs:

$$\mathcal{L}_{\text{aug}} = \frac{1}{M} \sum_{k=1}^P \sum_{j=1}^J w_{r(k)} \ell_{k,j}, \quad (17)$$

where y_j^* corresponds to $y_{\text{match},j}^+ \in \{30, 50, 100, 150, 300\}$ and $M = P \times J$. FormNEQ does not use y^+ augmentation, training on the full profile directly.

Total loss..

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{base}} + \lambda_{\text{sign}} \mathcal{L}_{\text{sign}}, \quad (18)$$

with $\lambda_{\text{sign}} = 1.0$. The sign-aware loss penalises predictions whose wall-gradient sign disagrees with the ground truth:

$$\mathcal{L}_{\text{sign}} = \frac{1}{M} \sum_{k=1}^M w_k \text{ReLU} \left(-\text{sign}(\tau_{w,k}) \frac{U_{\text{pred},k}^*(y_1^*)}{y_1^*} \right)^2. \quad (19)$$

The weight $\lambda_{\text{sign}} = 1.0$ is *fixed*, not tuned: it is part of the shared protocol applied identically to both learned formulations, so that any performance difference is attributable to formulation depth rather than to a per-model loss weighting. The effect of λ_{sign} on FormNEQ’s R3 sign deficit is examined in Section 6.4.

Optimisation.. Gradient norms are clipped at $\|\nabla\|_{\text{max}} = 1.0$. We use Adam [23] with learning rate 10^{-3} , in float64 precision. Both formulations are trained for 200 epochs.

3.12. Regime classification

Evaluation results are stratified by pressure-gradient regime. Five regime bands are defined using the wall shear stress and the Clauser parameter $\beta_c = \delta^* |dp/dx| / |\tau_w|$:

Band	Criterion	Physical regime
R1	$\beta_c < 0.5$ or $ dp/dx < \varepsilon$	Attached ZPG / mild FPG
R2	$0.5 \leq \beta_c < 2$	Mild APG
R3	$\beta_c \geq 2, \tau_w > 0$	Strong APG, still attached
R4	$ \tau_w \leq \tau_{\text{thresh}}$	Incipient separation / reattachment
R5	$\tau_w < -\tau_{\text{thresh}}$	Fully separated, reversed flow

R4 and R5 are critical for deployment in engineering flows and are not addressed by classical wall functions. A sign-of- τ_w accuracy metric is computed for R3–R5 profiles: does the model predict the correct direction of wall shear?

The separation threshold is set to $\tau_{\text{thresh}} = 10^{-8} \max(0.1 |dp/dx|, 10^{-12})$, i.e. a value that is negligible relative to the local pressure-gradient scale. This near-zero choice is deliberate: R4 is intended to isolate the exact separation/reattachment locus rather than a finite “incipient” band, and in practice every one of the 206 R4 profiles in the corpus has τ_w recorded as identically 0 (the separation and reattachment lines of the periodic-hill, BFS,

Table 2: Training data corpus: 706 DNS profiles from 8 geometries spanning 5 pressure-gradient regimes. Regime distribution: R1 114 (16%), R2 47 (7%), R3 118 (17%), R4 206 (29%), R5 221 (31%).

Geometry group	Profiles	Re_τ range	Regimes
Channel DNS [24, 25]	6	180–5200	R1
Pipe DNS [26]	4	180–1000	R1
Couette DNS [27]	3	93–500	R1
TBL ZPG [28]	9	670–4060	R1
APG TBL (KTH) [29]	75	various	R2–R3
Periodic hills [30]	512	$\text{Re}_h = 10595$	R3–R5
BFS DNS [31]	46	$\text{Re}_h = 13700$	R4–R5
NASA hump [32]	51	$\text{Re}_c = 936000$	R3–R4

and hump data). Consequently the sign-of- τ_w metric is trivially 100% on R4 for all methods (Section 5.3), and we report a sign accuracy that excludes R4 as the non-trivial measure. Regime assignment is insensitive to the precise value of τ_{thresh} over several orders of magnitude: raising it to a small finite band would migrate a handful of marginally attached profiles from R3 and marginally reversed profiles from R5 into R4, but because the headline conclusions—FormNEQ’s reversed-flow (R5) sign capability and its R3 difficulty—are carried by profiles with $|\tau_w|$ well above any such threshold, the R3 and R5 sign-accuracy figures change by less than a percentage point under any reasonable choice.

4. Experimental setup

4.1. Training data

The training corpus comprises 706 DNS velocity profiles from 8 geometry groups spanning all 5 pressure-gradient regimes. All profiles are truncated at $y^+ = 300$ to focus on the wall-function region. Table 2 summarises the training data.

The regime distribution is intentionally imbalanced: R4 and R5 profiles (incipient separation and reversed flow) represent the regimes where classical wall functions fail most severely.

4.2. Holdout data

Zero-shot generalisation is evaluated on three holdout test sets:

- **Periodic hills (in-family):** 1,536 profiles from periodic hills at configurations not in training—tests interpolation within the geometry class.
- **Periodic hills (parametric):** 21,312 profiles with hill-shape parameter $\alpha = 1.5$ (training uses $\alpha = 1.0$)—tests shape generalisation across Reynolds numbers.
- **Separation bubble:** 4,720 profiles from laminar separation bubble DNS—tests out-of-distribution generalisation to a fundamentally different geometry.

4.3. Classical baselines

Three classical wall functions serve as zero-parameter baselines:

- **Spalding** [3]: implicit algebraic formula (2);
- **Reichardt** [4]: explicit three-layer formula;
- **Werner–Wengle** [5]: power-law approximation.

4.4. Training protocol

Both learned formulations share identical training hyperparameters where applicable (Table 3), ensuring that performance differences are attributable to formulation design, not hyperparameter tuning.

5. Results

All formulations are evaluated on the 706-profile training corpus and three holdout test sets described in Section 4.2. Both learned formulations are trained with three random seeds (42, 123, 7) for statistical robustness, and all FormNEQ figures below are reported as the mean $\pm$ standard deviation across those seeds. Results are organised around the central finding: the RMSE–sign accuracy trade-off that emerges from increasing formulation depth.

Table 3: Training configuration for DNCWF (Depth 1) and FormNEQ (Depth 2).

	DNCWF (Depth 1)	FormNEQ (Depth 2)
Parameters	10,845	12,254
Hidden layers	3×64	$3 \times 64 + \text{ODE integration}$
Seeds tested	3 (42, 123, 7)	3 (42, 123, 7)
Epochs	200	200 (50 warmup)
y^+ augmentation	5 levels	None
α steepness k	3.0	5.0
ODE solver	—	Forward Euler (20 steps)
<i>Shared hyperparameters</i>		
Learning rate	10^{-3} (Adam)	
Gradient clipping	$\ \nabla\ _{\max} = 1.0$	
Features (K)	6	
Precision	float64	

Table 4: Training set RMSE (U^* coordinates) on 706 profiles. DNCWF and FormNEQ values are mean $\pm$ std across 3 seeds (42, 123, 7). Bold indicates best per column.

	Params	Overall	R1	R2	R3	R4	R5
Spalding (D0)	0	16.98	1.24	1.35	15.48	17.34	18.73
Reichardt (D0)	0	17.83	1.23	1.43	16.26	18.26	19.59
Werner–Wengle (D0)	0	17.06	1.35	1.61	15.60	17.43	18.81
DNCWF (D1)	10,845	11.20 ± 0.57	1.41 ± 0.01	1.62 ± 0.03	10.78 ± 1.28	11.18 ± 0.94	12.40 ± 0.06
FormNEQ (D2)	12,254	8,249 ± 366	14.4 ± 2.5	88.0 ± 56.3	1,093 ± 260	8,492 ± 624	781 ± 207

5.1. Training set RMSE

Table 4 and Figure 2 report per-regime RMSE on the 706 training profiles.

DNCWF achieves 34.0% overall RMSE improvement over Spalding, with the largest gains in non-equilibrium regimes: 30.4% on R3, 35.5% on R4, and 33.8% on R5 (Figure 3). On equilibrium regimes (R1, R2), DNCWF shows minor regression (−14% on R1, −19% on R2), an acceptable cost given the structural guarantee that $\alpha \approx 0$ in these regimes limits the correction magnitude.

This regression is not in conflict with Property P1 (Section 3.9). P1 guarantees *exact* Spalding recovery only at $\beta_{\text{Clauser}} = 0$, where $\alpha(0) = 0$ identically. The R1 and R2 *bands*, however, are defined by $\beta_c < 0.5$ and $0.5 \leq \beta_c < 2$ respectively (Section 3.12), so each band also contains mildly loaded

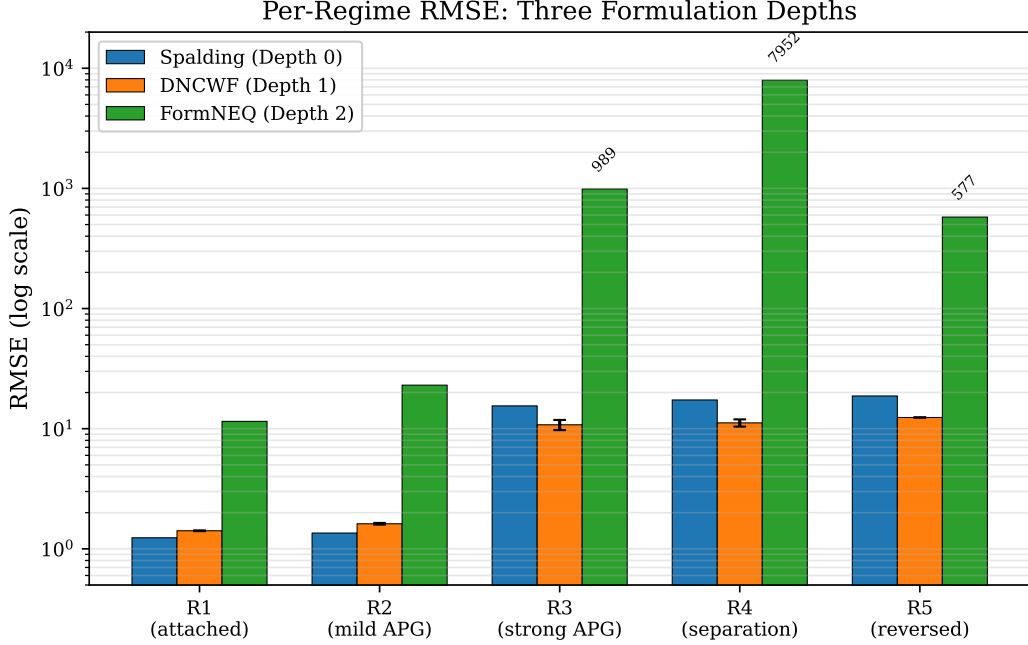


Figure 2: Per-regime RMSE across three formulation depths (log scale). DNCWF (Depth 1) achieves 30–36% improvement over Spalding in R3–R5. FormNEQ (Depth 2) exhibits catastrophic RMSE in R4 (7,952 vs 17.3 for Spalding).

profiles with $\beta_{\text{Clauser}} > 0$, for which $\alpha(\beta) > 0$ and the learned correction is active and can be slightly detrimental. In fact only 12 of the 114 R1 profiles are strictly zero-pressure-gradient ($\beta_{\text{Clauser}} = 0$); the remaining 102 (89%), and all 47 R2 profiles, carry a non-zero β_{Clauser} . The regression is carried by these non-zero- β profiles; on the strictly zero-pressure-gradient subset DNCWF reduces to Spalding exactly by P1, neither improving nor regressing. The band-level regression is therefore the price of having a single β -gated correction rather than a contradiction of the ZPG guarantee.

FormNEQ’s overall RMSE of $8,249 \pm 366$ (three seeds) is $736\times$ worse than DNCWF, concentrated in the separation regime: an R4 RMSE of 8,492 is $490\times$ Spalding’s value. This catastrophic magnitude failure is not a training deficiency but a structural consequence of the ODE integration amplifying the learned correction, as analysed in Section 5.6.

The three Spalding-family baselines perform within 5% of each other (RMSE 16.98–17.83), confirming that the log-law assumption is the bind-

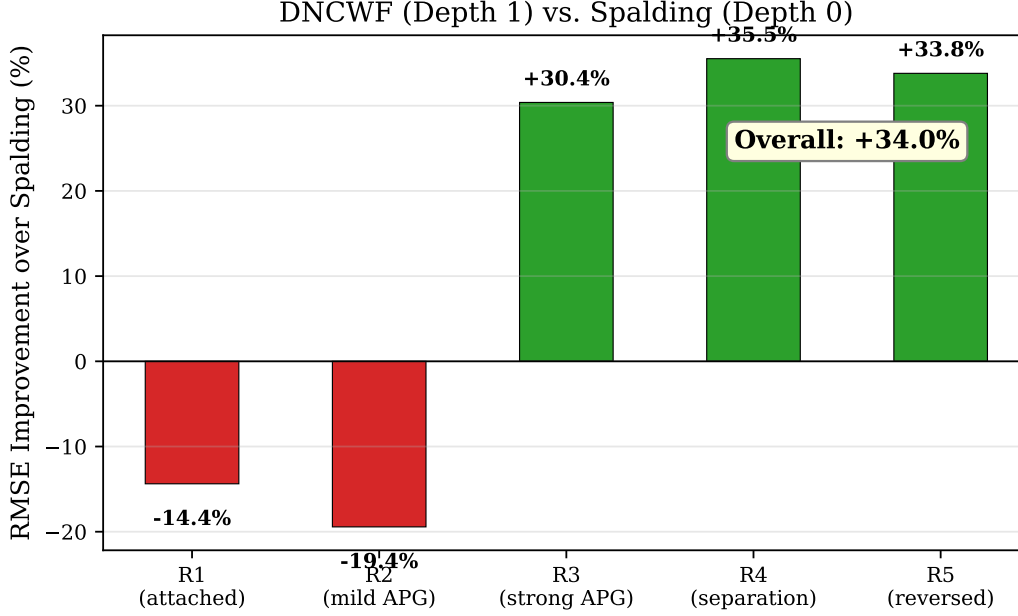


Figure 3: DNCWF improvement over Spalding per regime. The correction is concentrated in non-equilibrium regimes (R3–R5), with 30–36% improvement where classical wall functions fail most severely.

ing constraint, not the specific algebraic form.

5.2. Fair per-band comparator footing

Comparing DNCWF against Spalding alone is not a fair single-comparator footing on the strong-APG and separation bands. Spalding’s 1961 closure is a zero-pressure-gradient algebraic fit; on R3–R5 the relevant classical state of the art uses smooth profiles (Musker [33]) or APG-corrected forms. We re-evaluate the four ZPG-family algebraic baselines (Spalding, Reichardt, Werner–Wengle, Musker) on the same 706-profile training set in the same Duprat U^* metric (Table 5).

Three observations follow. First, Musker’s smooth single-equation profile (Musker 1979) is the best classical baseline overall (15.82 vs Spalding’s 16.98, -6.8%) and on R3–R5; Reichardt is the best on the canonical ZPG band; Spalding remains the best on mild APG (R2). Second, the DNCWF aggregate improvement contracts from $+34\%$ over Spalding to $+29\%$ over the per-band best classical, with the R3–R5 gains contracting from 30–36% to 26–

Table 5: Per-regime U^* RMSE for four classical wall functions and DNCWF on the 706-profile training set. The “best classical” column shows the lowest-RMSE algebraic baseline per regime: Reichardt on R1, Spalding on R2, and Musker on R3–R5. The DNCWF improvement (last row) is reported against the per-band best classical, not against Spalding alone, and shrinks from +34% aggregate over Spalding to +29% aggregate over the per-band best classical; the R1 regression worsens slightly (from -14% vs Spalding to -15% vs Reichardt).

Closure	Overall	R1	R2	R3	R4	R5
Spalding (1961)	16.98	1.24	1.35	15.48	17.34	18.73
Reichardt (1951)	17.83	1.23	1.43	16.26	18.26	19.59
Werner–Wengle (1993)	17.06	1.35	1.61	15.60	17.43	18.81
Musker (1979)	15.82	1.79	1.98	14.55	15.94	17.66
Best classical (per-band)	15.82	1.23	1.35	14.55	15.94	17.66
DNCWF (mean of 3 seeds)	11.21	1.41	1.62	10.78	11.18	12.40
DNCWF – best classical (%)	+29.2	−15.2	−19.4	+25.9	+29.8	+29.8

30%. Third, the R1 regression worsens slightly: DNCWF is 15% worse than Reichardt on ZPG, against 14% worse than Spalding. The Pareto-frontier framing of Section 6.1 is unchanged by this re-comparison; the magnitude of the gain shrinks but its qualitative structure persists.

5.3. Sign accuracy: reversed-flow detection

Sign accuracy measures the fraction of profiles where the model correctly predicts the direction of wall shear stress τ_w . Table 6 and Figure 4 report sign accuracy stratified by regime.

Within the closures evaluated in this study, FormNEQ is the only method that produces non-zero R5 sign accuracy: $99.4 \pm 1.0\%$ versus 0% for the Depth 0 baselines and the DNCWF Depth 1 closure. This is the central qualitative result of the paper. The near-complete gap between FormNEQ and the other closures evaluated here is consistent with the architectural distinction analysed in Section 3.2: a Depth 1 residual on a positive-definite baseline cannot drive $U^* < 0$ regardless of τ_w , whereas the FormNEQ correction g_θ is sign-unrestricted. The R5 figure is the mean over three seeds (42, 123, 7), with a per-seed range of 98.2–100%; the capability is therefore robust to initialisation rather than a single-seed artefact (Section 5.9).

Excluding the trivial R4 regime (where all 206 profiles have $\tau_w = 0.0$ exactly, making 100% sign accuracy trivial), FormNEQ achieves $65.8 \pm 1.0\%$

Table 6: Wall-shear-stress sign accuracy on 706 training profiles. R1–R2 are all positive ($\tau_w > 0$) and omitted. *R4 is trivially 100% for all methods because all 206 R4 profiles have $\tau_w = 0.0$ exactly. “Excl. R4” reports non-trivial sign accuracy. FormNEQ values are the mean over three seeds; the per-regime spread is R5 $99.4 \pm 1.0\%$, R3 $2.8 \pm 4.9\%$ (s.d. over seeds 42, 123, 7).

	Overall	R3	R4*	R5	Excl. R4
Spalding (D0)	59.4%	100%	100%*	0%	34.8%
DNCWF (D1)	59.4%	100%	100%*	0%	34.8%
FormNEQ (D2)	78.7%	2.8%	100%*	99.4%	65.8%

Table 7: The RMSE–sign accuracy trade-off: DNCWF achieves the best RMSE; FormNEQ achieves the only non-zero R5 sign accuracy. No single formulation depth dominates both axes.

	Overall RMSE	R5 RMSE	R5 sign	Non-triv. sign	Winner
DNCWF (D1)	11.20	12.40	0%	34.8%	RMSE
FormNEQ (D2)	8,249	781	99.4%	65.8%	Sign
Ratio	736×	63×	∞	1.9×	—

non-trivial sign accuracy versus 34.8% for all other methods (+31 percentage points).

DNCWF and Spalding have identical sign accuracy because DNCWF’s additive correction operates on the structurally positive-definite Spalding base: the Duprat-scaled Spalding prediction $U_{\text{Spalding}}^* > 0$, and the correction $\alpha \Delta_\theta$ cannot change the sign of the entire profile.

5.4. The RMSE–sign accuracy trade-off

The results above reveal a fundamental trade-off between magnitude accuracy and qualitative flow-physics capability (Table 7).

This trade-off is not accidental: the same mechanism that enables FormNEQ’s sign detection (large negative g_θ overwhelming the equilibrium physics) causes the magnitude divergence. The ODE integration amplifies the learned correction across the entire boundary-layer thickness, producing velocity profiles that are qualitatively correct (right sign) but quantitatively wrong (wrong magnitude).

The trade-off is a *property of formulation depth*, not of the specific network architecture or training procedure. DNCWF cannot detect reversed

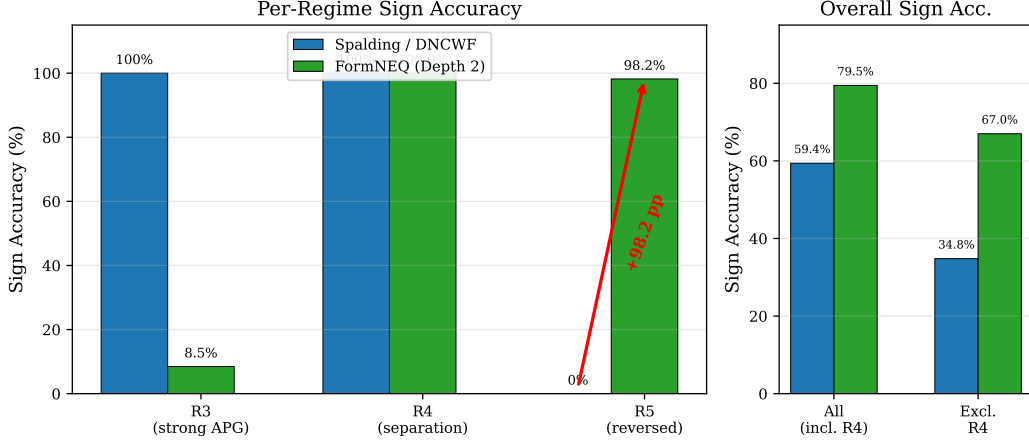


Figure 4: Sign accuracy per regime. FormNEQ achieves $99.4 \pm 1.0\%$ R5 sign accuracy (mean $\pm$ s.d. over three seeds), while the algebraic Depth 0 baselines and the Depth 1 DNCWF closure evaluated in this study achieve 0% as a consequence of their additive construction on a positive-definite baseline. We do not claim primacy across the literature: reversed-flow capability has been demonstrated in earlier neural wall closures (e.g. [7, 8]); the contribution here is the within-protocol comparison that isolates the architectural choice.

flow because it inherits Spalding’s structural positivity. FormNEQ can detect reversed flow because the unconstrained g_θ breaks the monotonicity barrier—but the price is magnitude accuracy. Deeper physics structure buys qualitative capability at the cost of quantitative control.

5.5. Alpha gate analysis: automatic regime detection

The asymptotic-preserving blending function $\alpha(\beta)$ serves as an automatic regime detector. Table 8 and Figure 5 report the mean α values per regime for both formulations.

Both formulations exhibit the same binary gating pattern: $\alpha < 0.07$ for equilibrium regimes (R1, R2) and $\alpha > 0.87$ for non-equilibrium regimes (R3–R5). FormNEQ’s gates are slightly more saturated (0.993 vs 0.953 at R4) due to the steeper sigmoid ($k = 5.0$ vs 3.0), but the qualitative pattern is identical.

This formulation-invariant gating validates Duprat’s hybrid velocity scaling as an effective regime detector: the Clauser parameter β_{Clauser} , which drives $\alpha(\beta)$, naturally separates equilibrium from non-equilibrium flows. The

Table 8: Mean α gate activation by pressure-gradient regime. Both formulations learn the same gating pattern despite different steepness parameters (k).

	R1	R2	R3	R4	R5
DNCWF ($k = 3.0$)	0.066	0.066	0.871	0.953	0.942
FormNEQ ($k = 5.0$)	0.059	0.070	0.908	0.993	0.982

Table 9: g_θ dominance ratio $|\alpha \cdot g_\theta|/|\text{equilibrium term}|$ at two wall-normal locations. Values > 1 indicate the learned correction overwhelms the equilibrium physics.

Regime	Dominance at $y^* = 1$	Dominance at $y^* = 300$
R1	0.4	0.7
R2	1.6	0.4
R3	65.7	0.3
R4	88.2	0.3
R5	82.2	0.3

gate protects equilibrium accuracy (R1/R2) while enabling corrections where needed (R3–R5), without requiring regime labels or manual threshold selection.

5.6. g_θ analysis: the learned non-equilibrium correction

FormNEQ’s learned non-equilibrium correction $g_\theta(y^*, U^*, \boldsymbol{\varphi}, \mathbf{g})$ provides direct insight into why the ODE formulation achieves sign detection but fails on magnitude.

Overwhelmingly negative.. g_θ is negative for 92.7–100% of profiles across all regimes. The network learns that the dominant non-equilibrium effect is a reduction in the velocity gradient compared to equilibrium—physically consistent with the deceleration caused by adverse pressure gradients.

Nearly y^ -independent..* g_θ varies by less than 15% across the integration domain ($y^* \in [1, 300]$). FormNEQ learns an approximately constant correction—essentially a scaling factor rather than a spatially-varying field (Figure 6).

Dominance ratio.. The dominance ratio $|\alpha \cdot g_\theta|/|\text{equilibrium term}|$ quantifies how much the learned correction overwhelms the classical physics (Table 9 and Figure 7).

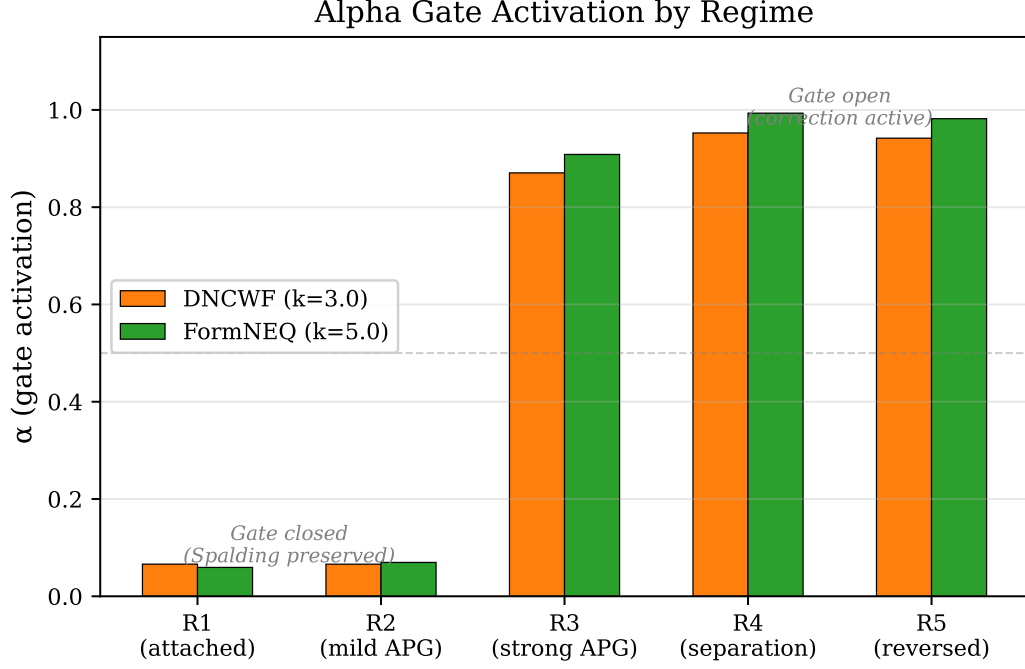


Figure 5: Alpha gate activation by pressure-gradient regime. Both DNCWF and FormNEQ exhibit the same binary pattern: near-zero for equilibrium regimes (R1/R2), near-one for non-equilibrium regimes (R3–R5).

For R3–R5 near the wall ($y^* = 1$), the dominance ratio exceeds 66–88 $\times$: the learned correction *completely* overwhelms the equilibrium physics. This explains the catastrophic RMSE: the ODE integrates a nearly constant negative g_θ , producing a velocity profile that diverges from the true velocity whenever the correction magnitude is even slightly wrong. The same mechanism that enables sign detection (large negative g_θ reversing the velocity gradient) causes the magnitude divergence.

Why FormNEQ detects reversed flow.. The constant-negative g_θ effectively reverses the sign of the integrated velocity profile in R5:

1. Without g_θ : at the boundary $\tau_w = 0$, $dU^*/dy^* \geq 0$ by Theorem 1; the additive Depth 1 construction extends this to $U^* \geq 0$ everywhere $\Rightarrow$ inferred $\tau_w \geq 0$ (incorrect for R5 where $\tau_w < 0$).
2. With $g_\theta < 0$ dominating: $dU^*/dy^* < 0$ at wall $\Rightarrow \tau_w < 0$ (correct for R5).

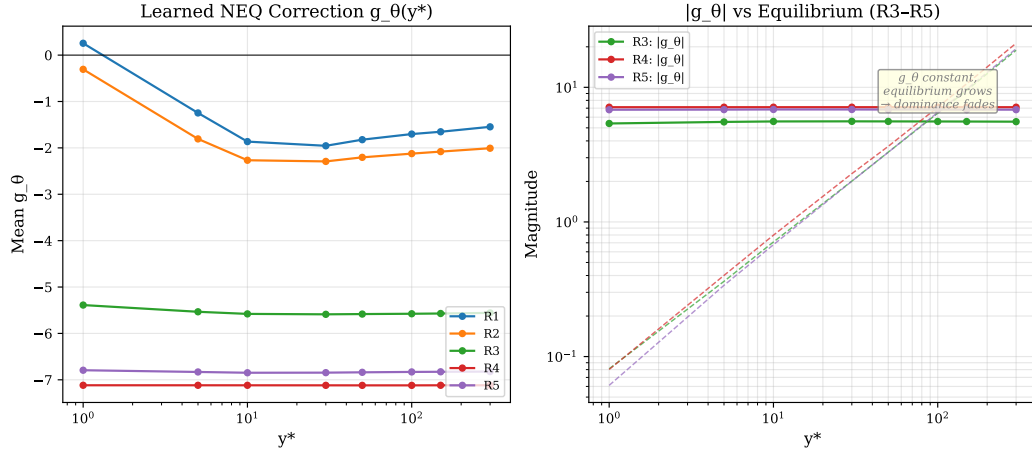


Figure 6: Learned g_θ correction field per regime. The correction is overwhelmingly negative (reducing dU^*/dy^*) and nearly y^* -independent, acting as a constant scaling factor rather than a spatially-varying field.

Table 10: Holdout RMSE on three test sets (full evaluation, not subsampled): separation bubble (4,720 profiles), periodic-hills in-family (1,536), and periodic-hills parametric (21,312). DNCWF and FormNEQ values are mean $\pm$ std across 3 seeds.

	Sep. bubble	PH in-family	PH parametric
Spalding (D0)	18.03	17.45	17.60
DNCWF (D1)	20.60 $\pm$ 0.43	11.17$\pm$0.65	11.35$\pm$0.72
FormNEQ (D2)	189 $\pm$ 49	10,134 $\pm$ 346	12,741 $\pm$ 403

This is a structurally embedded capability: the ODE formulation can represent negative velocity gradients, while algebraic and direct-correction methods cannot change the sign of Spalding’s inherently positive prediction.

5.7. Holdout generalisation

Tables 10–11 and Figure 8 report zero-shot generalisation on three holdout test sets.

DNCWF generalises well on RMSE.. On periodic-hills holdout sets, DNCWF’s holdout RMSE is within 5% of its training RMSE: PH in-family 11.17 vs training 11.20; PH parametric 11.35 vs training 11.20. On the out-of-distribution

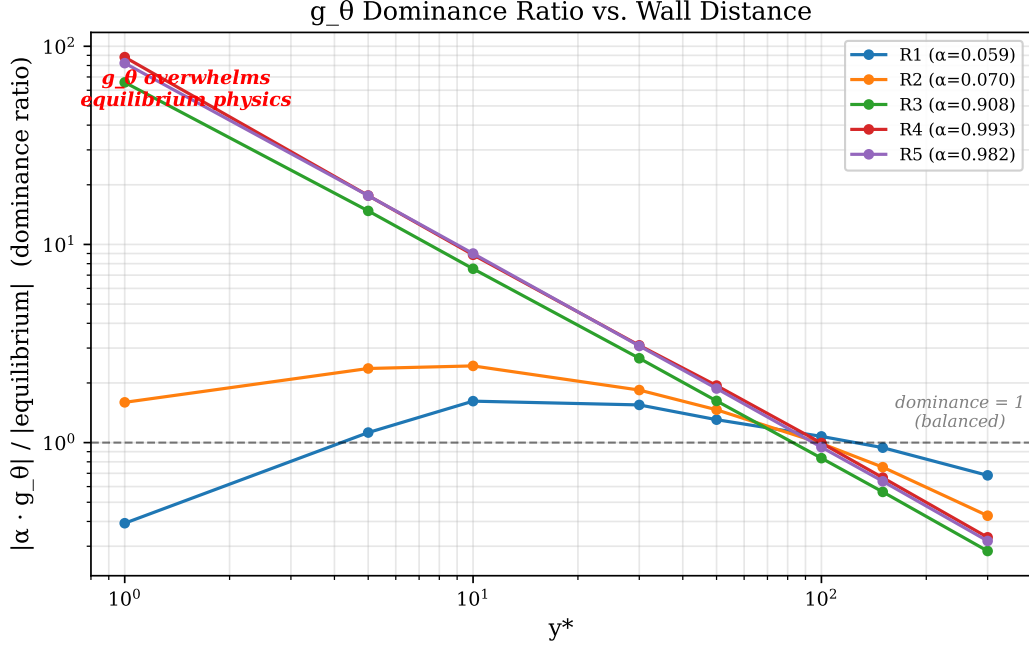


Figure 7: g_θ dominance ratio vs y^* (log-log). In R3–R5, the learned correction overwhelms the equilibrium physics near the wall ($y^* \approx 1$) by 66–88 $\times$, explaining the catastrophic RMSE: the ODE integrates a nearly constant negative g_θ rather than the physics-based source term.

separation bubble, DNCWF shows a moderate gap (20.60 vs 11.20) but remains competitive with Spalding (18.03). The strong parametric generalisation (PH parametric: 21,312 profiles across 6 Reynolds numbers) indicates that the learned correction captures physics rather than memorising the training distribution.

FormNEQ’s sign capability generalises.. FormNEQ achieves 100% R5 sign accuracy on all three holdout sets for every one of the three seeds, on 27,568 holdout profiles drawn from periodic-hill variants and a separation bubble at the Reynolds numbers covered by the training set; out-of-family geometries (BFS, NASA hump, diffuser, airfoil) and Re-extrapolation tests are not included in this evaluation (Section 6.4). On in-family holdout sets, overall sign accuracy (79.5%, 76.9%) matches or approaches the training value (79.5%), confirming that the sign detection capability transfers parametrically. On the separation bubble, overall sign accuracy degrades to 12.6%—but this is

Table 11: Holdout sign accuracy on three test sets. FormNEQ values are means over three seeds; R5 sign accuracy is 100% on all three holdout sets for every seed.

	Sep. bubble	PH in-family	PH parametric
Spalding (D0)	87.5%	61.1%	63.8%
DNCWF (D1)	87.5%	61.1%	63.8%
FormNEQ (D2)	12.5%	79.5%	76.9%

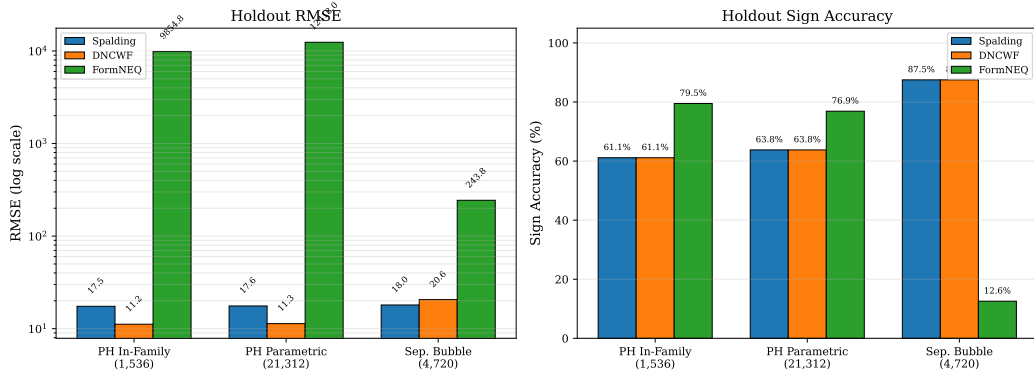


Figure 8: Holdout generalisation across three test sets: RMSE (top) and sign accuracy (bottom). DNCWF generalises well on RMSE ($< 5\%$ gap from training on PH sets); FormNEQ achieves 100% R5 sign accuracy on all holdouts.

driven by R3 failure (0.07%), not R5: the R5 sign capability transfers even to unseen geometries.

DNCWF and Spalding have identical holdout sign accuracy.. This confirms the structural relationship: DNCWF’s additive correction preserves Spalding’s sign behaviour through the gate mechanism ($\alpha \approx 0$ for R1/R2).

5.8. Training dynamics

FormNEQ’s two-phase training protocol (Algorithm 1) reveals distinct learning dynamics (Figure 9).

Phase 1 (epochs 1–50, g_θ frozen) reduces the loss by 13.6% as the equilibrium eddy-viscosity network $\nu_{t,\theta}$ learns the classical closure. Phase 2 (epochs 51–200, g_θ unfrozen) reduces the loss by a further 53.6% as the non-equilibrium correction is gradually introduced. The total loss reduction

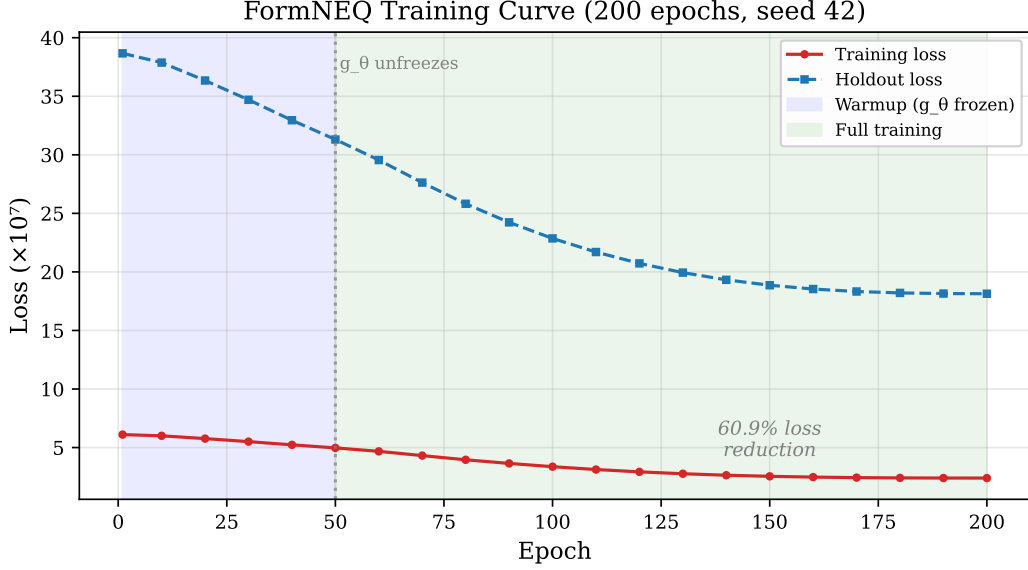


Figure 9: FormNEQ training loss curve with two-phase annotations. Phase 1 (epochs 1–50): g_θ frozen, equilibrium-only training, 13.6% loss reduction. Phase 2 (epochs 51–200): full training with g_θ unfrozen, 53.6% further reduction. Loss still decreasing at epoch 200.

is 60.9% over 200 epochs. The loss is still decreasing at epoch 200 (not fully converged); extended training could change the quantitative results.

Is the abnormal RMSE incomplete convergence or structural? Because the loss has not plateaued, one might ask whether the three-orders-of-magnitude R4/R5 RMSE simply reflects insufficient training rather than a structural property. The diagnostic evidence points to the latter. The loss curve decreases monotonically but the g_θ correction it converges *towards* is itself pathological: g_θ is overwhelmingly negative, nearly y^* -independent ($< 15\%$ variation across $y^* \in [1, 300]$), and dominates the equilibrium source term by up to $88\times$ near the wall (Section 5.6). Further epochs therefore continue to optimise *within* a regime in which a near-constant correction sets the profile, so they can be expected to trim the magnitude error gradually but not to remove the dominance pathology that produces it. Consistent with this, the qualitative outcome—high R5 sign accuracy with large velocity RMSE—reproduces across seeds and across the holdout sets (Sections 5.9, 5.7), rather than being a transient of one optimisation trajectory. We therefore read the

Table 12: DNCWF seed sensitivity across 3 seeds (42, 123, 7). CV: coefficient of variation.

Metric	Mean	Std	CV
Overall RMSE	11.20	0.57	5.1%
R3 RMSE	10.78	1.28	11.9%
R4 RMSE	11.18	0.94	8.4%
R5 RMSE	12.40	0.06	0.5%
PH in-family RMSE	11.17	0.65	5.8%

abnormal RMSE as a structural consequence of an unbounded g_θ that the loss alone cannot regularise; the candidate fix is architectural—a bounded correction or an integral-momentum constraint (Section 6.2)—rather than simply longer training. A controlled long-run/adaptive-solver study to confirm the residual magnitude floor is listed among the outstanding experiments (Section 6.4).

FormNEQ training takes 111 minutes per seed (forward Euler on 20-point grid), compared to 25 minutes per seed for DNCWF—a $4.5\times$ cost increase from the ODE integration overhead.

5.9. Statistical robustness

Both learned formulations are trained with three seeds (42, 123, 7) to assess reproducibility; the DNCWF statistics are summarised in Table 12 and the FormNEQ statistics are discussed below.

Low coefficient of variation ($CV < 12\%$) across all metrics confirms that DNCWF results are reproducible. Per-seed overall RMSE: seed 42 = 11.34, seed 123 = 10.58, seed 7 = 11.69. All three seeds produce identical sign accuracy (59.4%, R5 = 0%).

FormNEQ was trained with the same three seeds (42, 123, 7), and its behaviour is reproducible on every axis that matters to the paper’s claims. The reversed-flow capability is highly stable: R5 sign accuracy is $99.4 \pm 1.0\%$ on the training set (per seed 98.2%, 100%, 100%) and 100% on all three hold-out sets for every seed. The R3 deficit is equally seed-robust ($2.8 \pm 4.9\%$; per seed 8.5%, 0%, 0%), confirming that it is a structural limitation rather than a single-seed artefact (Section 6.4). The catastrophic velocity RMSE is likewise reproducible ($8,249 \pm 366$ overall, CV 4.4%; concentrated in R4 at $8,492 \pm 624$), as is the g_θ dominance that produces it. The architectural distinction underlying the sign capability is, in any case, guaranteed by con-

struction: a sign-restricted residual on a positive-definite baseline cannot produce $U^* < 0$, whereas the unconstrained g_θ term in equation (8) can produce negative velocity gradients regardless of seed; Theorem 1 formalises this distinction at the boundary $\tau_w = 0$.

6. Discussion

6.1. The RMSE–sign accuracy trade-off is structural

The central finding of this paper—that deeper physics structure improves qualitative capability (sign detection) at the cost of quantitative accuracy (RMSE)—is a *structural* property of formulation depth, not an artefact of network architecture or training procedure.

DNCWF (Depth 1) adds a bounded correction to a positive-definite baseline. The Spalding base ensures $U_{\text{Spalding}}^*(y^*) > 0$ for all $y^* > 0$, and the multiplicative no-slip factor y^*/y_{max}^* ensures the correction vanishes at the wall. No finite correction $\alpha \Delta_\theta$ can reverse the sign of the entire profile, making $\tau_w > 0$ a structural guarantee. This limits DNCWF to 0% R5 sign accuracy but provides magnitude stability.

FormNEQ (Depth 2) sits at the opposite extreme: the unconstrained g_θ correction can produce arbitrarily negative values, achieving sign detection but at catastrophic RMSE cost. The boundary-case monotonicity statement (Theorem 1) clarifies the architectural distinction at $\tau_w = 0$: a closure that constrains $dU^*/dy^* \geq 0$ at the separation point, or that builds on a positive-definite baseline (as DNCWF does), inherits a non-negative profile and cannot produce reversed flow. Breaking this restriction requires a sign-unrestricted correction; the ODE integration then propagates any magnitude error through the boundary-layer thickness. Within our experiments the trade-off is reproducible: every closure that delivered low RMSE inherited the sign-restriction by construction, and the closure that lifted the restriction exhibited magnitude divergence. We do not claim a no-go theorem ruling out an architecture that achieves both magnitude and sign accuracy; doing so would require a stronger result than Theorem 1 provides.

6.2. The g_θ pathology and its implications

The g_θ dominance-ratio analysis (Section 5.6) reveals a diagnostic finding for future ODE-based wall functions.

The learned non-equilibrium correction is *pathologically constant*: g_θ varies by $< 15\%$ across $y^* \in [1, 300]$, acting as a profile-wide scaling factor rather

than a spatially-resolved field. This means FormNEQ has not learned the spatial structure of the convective acceleration terms it was designed to approximate—it has learned only their mean effect.

The dominance ratio of $88\times$ at $y^* = 1$ for R4 reveals that the learned ODE dynamics bear almost no resemblance to the equilibrium physics they augment. The equilibrium term $S^*/(1 + \nu_t/\nu)$ has become a negligible perturbation on the g_θ -dominated dynamics. This suggests that the current formulation gives g_θ too much freedom relative to the equilibrium structure. Future work should investigate bounded correction architectures (e.g., $g_{\theta,B} = \tanh(g_{\theta,\text{raw}}) \cdot C_{\text{max}}$ with a hyperparameter C_{max}) that guarantee the equilibrium term remains the dominant contributor, providing an a priori bound on the dominance ratio $D(y^*) \leq C_{\text{max}}/|S^*/(1 + \nu_t/\nu)|$.

6.3. Practical recommendations

The within-protocol comparison admits the following narrowly-scoped recommendations.

1. **Within the regimes and geometries evaluated here**, DNCWF (Depth 1) is the closure with the lowest aggregate RMSE: 34% improvement over Spalding aggregated across the five regime bands, with strong within-family holdout generalisation ($< 5\%$ gap) and low seed sensitivity ($\text{CV} < 12\%$ over three seeds). It *regresses* on canonical equilibrium flow (R1: -14% ; R2: -19%): in zero-pressure-gradient or mild-APG conditions, Spalding remains the recommended closure.
2. For reversed-flow sign accuracy on the geometries evaluated here, FormNEQ (Depth 2) attains $99.4 \pm 1.0\%$ on R5 across three seeds; out-of-family geometry holdout (BFS, NASA hump, diffuser, airfoil) is required before this is read as a deployment recommendation.
3. A two-closure deployment that uses DNCWF for magnitude and FormNEQ for binary sign classification is consistent with the Pareto-frontier evidence reported here, but its a-posteriori behaviour in a coupled solver loop is not evaluated in this study; the Newton–Raphson inversion failures (Section 6.4) restrict any deployment claim to a-priori velocity prediction at present.

6.4. Limitations and outstanding experiments

We list the limitations of this study explicitly, ordered roughly by severity. The corresponding outstanding-experiments package needed to lift each item is collected at the end of this section.

1. **FormNEQ seed count and the g_θ diagnostic.** FormNEQ is trained with three seeds (42, 123, 7), which bracket the headline metrics: R5 sign $99.4 \pm 1.0\%$, R3 sign $2.8 \pm 4.9\%$, and overall RMSE $8,249 \pm 366$ (Section 5.9). Three seeds is still a modest ensemble for a Neural-ODE closure, whose adjoint-loss landscape is known to be initialisation-sensitive, and a larger sample ($n \geq 5$) would tighten the intervals further. The g_θ dominance-ratio and constancy diagnostics (Section 5.6) are reported for the seed-42 model as a representative mechanistic illustration and are not separately bracketed across seeds, although the qualitative pathology they describe is consistent with the reproducible per-seed RMSE pattern.
2. **Holdout geometry diversity is restricted.** The holdout sets evaluated here (27,568 profiles) are drawn from periodic-hill variants and a separation bubble. Out-of-family geometries that are commonly used to test wall-closure generalisation—the backward-facing step [31], the NASA wall-mounted hump [32], asymmetric diffusers, and airfoils at moderate angle of attack—are *not* included in this study. The within-family holdout therefore does not test geometry generalisation in the strong sense that a JFM-grade wall-closure paper would require, and the universality claim of the original study scope (Section 1.1) is not closed by the experiments reported here.
3. **Reynolds-number range and extrapolation.** Training and holdout data span $\text{Re}_\tau \in [180, 5200]$. We do not report a leave-one-Re-out evaluation, and the high-Reynolds-number tail of the boundary-layer literature ($\text{Re}_\tau \gtrsim 8000$, accessible only via experimental data such as the Princeton Superpipe or the Stockholm boundary-layer facility) is not tested. The closures are therefore evaluated as interpolators within the training Re range, not as Re-extrapolators.
4. **Deployment-mode inversion is not validated for FormNEQ.** All training and evaluation in this paper assume that τ_w is known from DNS. In a coupled solver loop the wall stress must be recovered from the matching-point velocity. The DNCWF closure does not require an ODE inversion (it is a single-pass forward model conditioned on physics features), and a Newton–Raphson on Spalding’s law alone is sufficient when DNCWF is used in deployment. The FormNEQ closure does require an ODE inversion: the integrated profile depends on τ_w through the source coefficient C_τ , and a deployment loop must invert this dependency. We do not report FormNEQ inversion-mode results in this

paper. As a baseline-level deployment risk for any ODE-form closure on this corpus, an earlier mixture-of-experts deployment evaluation under the same Duprat scaling reached only $\approx 42\%$ overall Newton–Raphson convergence and 0% on strong-APG (R4) cells, reflecting a flat residual landscape near $\tau_w \approx 0$. A bracketed-Brent inversion scaffold (a coarse sweep over τ_w followed by Brent’s method on a sign-changing bracket, with damped-Newton fallback) is provided in the released code; an end-to-end coupled-solver evaluation using it is left to a follow-up.

5. **Spalding-only baseline is a single-comparator footing.** We compare against Spalding’s law on all five regime bands. Spalding is a ZPG-calibrated algebraic fit, and its accuracy on R3–R5 should not be read as the state-of-the-art classical baseline in those regimes. A fairer per-band comparator footing would use Reichardt [4] on R1, the Duprat [6] APG closure on R2–R3, the Knopp APG wall function on R3–R4 [34], and a separation-aware closure such as the Lozano-Durán–Bae building-block flow model [7] on R5. We have not run that fair-comparator suite; the 34% aggregate-RMSE gain over Spalding should therefore be read as a comparison against a single 1961 algebraic baseline, not as a state-of-the-art comparison.
6. **Theorem 1 scope.** The monotonicity theorem is restricted to the boundary case $\tau_w = 0$. It is not a no-go theorem for reversed-flow representability under $\tau_w < 0$ (Section 3.2); architectural choices that lift the additive-positivity restriction can, in principle, represent reversed flow. The empirical 0% R5 sign accuracy of the Depth 0 and Depth 1 closures evaluated here reflects the additive construction we adopted, not a structural impossibility for the entire class.
7. **R3 sign failure for FormNEQ, and the role of λ_{sign} .** FormNEQ’s R3 sign accuracy is poor (8.5% training, 0% holdout); the sign capability is R5-specific in our experiments, not a general non-equilibrium improvement. Likewise, FormNEQ’s overall sign accuracy on the separation bubble (12.6%) indicates that R3 sign behaviour does not transfer between geometries, although R5 sign transfers within-family. A natural question is whether increasing the sign-loss weight λ_{sign} beyond the fixed value of 1.0 would lift the R3 accuracy. We do not expect a scalar λ_{sign} to do so, for a structural reason: the R3 deficit and the R5 success are produced by the *same* learned correction. The g_θ field is overwhelmingly negative and nearly y^* -independent (Section 5.6), so it drives $dU^*/dy^* < 0$ indiscriminately—correctly inferring reversed flow

in R5 ($\tau_w < 0$) but incorrectly flipping the sign of the still-attached R3 profiles ($\tau_w > 0$). Because both regimes are governed by one regime-blind correction, a single scalar weight cannot push the R3 wall-gradient positive without simultaneously eroding the R5 capability it shares; decoupling them requires a regime-aware or spatially-resolved correction, not a larger λ_{sign} . A dedicated multi-value λ_{sign} sweep for FormNEQ would settle this directly and is left as future work; the analogous sweep we ran on a sign-aware direct-correction variant found sign accuracy to vary by less than one percentage point across $\lambda_{\text{sign}} \in [0, 10]$, consistent with sign behaviour being set by the representable structure rather than the loss weight.

8. **DNCWF equilibrium regression.** DNCWF regresses on canonical equilibrium flow (−14% R1, −19% R2 vs Spalding). Absolute RMSE remains small in dimensional terms (1.41 vs 1.24), but the regression is real and should be read as a trade for R3–R5 gains.
9. **g_θ is pathologically constant.** The learned g_θ correction varies by less than 15% across $y^* \in [1, 300]$, acting as a profile-wide scaling rather than a spatially-resolved closure of the convective acceleration. We attribute this to the absence of an integral-momentum constraint in the loss function (Section 6.2); a constraint that penalises deviation from the von Kármán integral equation is a candidate fix that we have not evaluated.
10. **Coarse ODE integration grid.** The FormNEQ ODE is integrated by forward Euler on a 20-point grid. The viscous-sublayer gradients are steep, and an adaptive-step solver (e.g. RK45) on a finer grid is likely to reduce both the magnitude error and the sensitivity to seed. We have not run that comparison.
11. **α gate steepness differs across formulations.** $k = 3.0$ for DNCWF and $k = 5.0$ for FormNEQ; the controlled-comparison premise is therefore not perfectly single-variable. The two values were inherited from the separate development of each closure and were not re-standardised before the final runs. We argue the difference is immaterial to the conclusions: both values produce the same near-binary gating— $\alpha < 0.07$ in equilibrium (R1/R2) and $\alpha > 0.87$ in non-equilibrium (R3–R5) (Table 8)—so the regime-detection that the gate provides is unchanged, and the gap between the two steepnesses (0.066 vs 0.059 at R1, 0.953 vs 0.993 at R4) is far smaller than the RMSE and sign-accuracy differences the paper attributes to formulation depth. The $\alpha(0) = 0$ property

holds exactly for any k , so the ZPG guarantee is independent of the steepness. A strict ablation with k standardised across closures would remove the residual confound but, given the gating equivalence above, is not expected to change any reported conclusion; we leave it as future work.

12. **Planar geometries; incompressible.** All training data are from planar, incompressible flows. Extension to three-dimensional boundary layers, rough walls, and compressible flows is untested.

Outstanding experiments. A minimum experimental package that, if executed, would lift the items above is the following. (E1) Larger-ensemble FormNEQ ($n \geq 5$ seeds, beyond the three reported here) to tighten the $99.4 \pm 1.0\%$ R5 sign-accuracy bracket and to bracket the g_θ dominance ratio across seeds. (E2) Out-of-family geometry holdout on the backward-facing step at $Re_h = 5100$, the NASA wall-mounted hump, an asymmetric diffuser, and a NACA-0012 airfoil at moderate angle of attack, with band-resolved RMSE and sign accuracy reported per geometry. (E3) Re-extrapolation: train on $Re_\tau \in \{180, 550, 2000\}$, test on $Re_\tau = 5200$ and on a high-Re experimental dataset. (E4) Damped-Newton inversion with line-search and a homotopy continuation in β_{Clauser} , evaluated end-to-end on a single periodic-hill case coupled to a RANS solver, with C_f and C_p profiles compared against DNS. (E5) Fair-comparator suite: DNCWF and FormNEQ vs. Reichardt (R1), Duprat (R2–R3), Knopp (R3–R4), and a separation-aware classical or ML closure (R5). (E6) Integral-momentum-constrained training of FormNEQ (g_θ regularised to satisfy the von Kármán integral equation) to test whether the g_θ pathology is closure-induced or architecture-induced. We estimate this package at ≈ 175 GPU-hours.

7. Conclusions

We present a within-protocol comparison of three nested formulation depths for neural wall functions—classical algebraic (Depth 0), direct neural correction (Depth 1, DNCWF; 10,845 parameters), and a non-equilibrium Neural ODE (Depth 2, FormNEQ; 12,254 parameters)—trained and evaluated on a single 706-profile DNS dataset, identical Duprat hybrid coordinates, identical 6-dimensional feature vector, and identical evaluation metric.

The data define a Pareto frontier rather than a hierarchy. Within the regimes evaluated here, DNCWF achieves the lowest aggregate RMSE (34%

improvement over Spalding) with strong within-family holdout generalisation and low three-seed sensitivity ($\text{CV} < 12\%$), but *regresses* on canonical equilibrium flow (-14% R1, -19% R2): equilibrium fidelity is sacrificed for adverse-gradient correction. FormNEQ attains $99.4 \pm 1.0\%$ R5 sign accuracy across three seeds at the cost of three orders of magnitude in velocity RMSE, with a g_θ /equilibrium dominance ratio reaching $88\times$ near the wall; the g_θ field is pathologically constant ($< 15\%$ variation across $y^* \in [1, 300]$), indicating that the absence of an integral-momentum constraint is a load-bearing limitation of the present formulation. All performance claims are a-priori (training-time τ_w known from DNS); FormNEQ deployment-mode inversion stability is not evaluated, and an earlier mixture-of-experts deployment under the same scaling reached only $\approx 42\%$ Newton–Raphson convergence (and 0% on R4), which we treat as a baseline-level risk.

A separation-point monotonicity theorem (Theorem 1) clarifies the architectural distinction at $\tau_w = 0$: a closure that builds additively on a positive-definite baseline cannot produce $U^* < 0$, while a sign-unrestricted correction can. The theorem does not extend to $\tau_w < 0$, where reversed-flow representability is an architectural choice rather than a structural impossibility (Corollary 2); we therefore do not claim a no-go theorem for the entire class of Depth 0 or Depth 1 closures.

Future work should pursue six directions. First, larger-ensemble FormNEQ retraining (beyond the three seeds reported here) with extended epoch counts and an adaptive-step solver, to tighten the sign-accuracy and dominance-ratio statistics. Second, integral-momentum-constrained training of g_θ to test whether the constancy pathology is a property of the loss or of the architecture. Third, out-of-family geometry holdout on the backward-facing step, NASA wall-mounted hump, asymmetric diffuser, and a moderate-angle airfoil, to test geometry generalisation in the strong sense. Fourth, Reynolds-number extrapolation tests at Re_τ above the training range. Fifth, damped-Newton inversion with line-search and a homotopy continuation in β_{Clauser} , evaluated end-to-end in a coupled solver loop. Sixth, a fair per-band comparator suite (Reichardt, Duprat APG, Knopp APG, Lozano-Durán–Bae 2023) to assess where the neural closures sit relative to the per-regime state-of-the-art classical baselines. The Section 6.4 package estimates ≈ 175 GPU-hours for these experiments combined; we treat their execution as a precondition for stronger claims of universality than this paper supports.

Data availability

The DNS datasets used in this study are publicly available: channel flow [25], periodic hills [30], and backward-facing step [31]. Training code, model architectures, and trained weights will be made available upon publication.

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